

Fazul Al Rehman

Offensive Security Researcher | Bug Bounty Hunter

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EDUCATION

BS Computer Science | Institute of Business Administration (IBA), Karachi

2021 – 2025

HSSC (Pre-Engineering) | Superior College of Sciences

2019

PROFESSIONAL EXPERIENCE

Independent Security Researcher | Offensive Security & Bug Bounty

2019 – Present

Full-time bug bounty hunter since Jan 2026. Offensive security research and responsible disclosure across enterprise, e-commerce, gaming, and Web3 targets.

Samsung Pass, Critical Production Vulnerability

Critical, Bugcrowd

- Broke access control to reach internal environment secrets, including a Jenkins CI/CD Slack token with admin-level access to sensitive operational channels.
- Mapped live dashboards, Terraform, SSH key rotations and internal credentials, and gained further controls leading directly to Samsung's internal databases and servers; disclosed via Bugcrowd for a **\$10k** bounty on a biometric platform affecting 100M+ users.

Walmart Marketplace, Payment Access-Control Bypass

Critical

- Critical broken-access-control flaw in Walmart Marketplace Seller Central: any authenticated account could reroute incoming funds on any merchant account, leaving all 100k+ Marketplace merchants directly drainable, an estimated **\$500M+** in payment flow at risk.

Large-Scale Credential Exposure via Dorking

Disclosure in progress

- Systematic search-engine dorking surfaced 20,000+ exposed private keys and API secrets across AWS, Stripe, Twilio, Razorpay, OpenAI, Anthropic, Shopify, Twitch, and GitHub.
- Aggregated GitHub keys exposed auto-deploying access to 1,600+ private repositories; Shopify secret keys affected multiple e-commerce brands (Shark Tank India-featured startups, ushopmall, Greyson Clothiers). Coordinating responsible disclosure with affected organizations.

IBA Karachi, University Portal Security Assessment

Responsibly disclosed

- Identified multiple critical vulnerabilities across the portals of IBA Karachi, a top Pakistani university, and disclosed them to administration, prompting a full migration to a new portal now in production.
- Findings: a database exposure leaking confidential personal records (including stored .png files), and an authentication bypass reaching faculty accounts up to super-admin level.

EA Origin, Security-Question Authentication Bypass

Auth bypass

- Security-question bypass in EA Origin's account-recovery flow, enabling authentication without the legitimate answers.
- Impact: every account on the platform was exposed to takeover and theft, since the recovery flow was the only barrier standing between an attacker and full account access.

Dell, Invoice-PDF Authorization Bypass

Auth bypass

- Authorization bypass on Dell allowing retrieval of invoice PDFs through serial-number enumeration.
- Retrieved invoices exposed real customer names, order details and account information, exactly the kind of detail that could be used to craft convincing social-engineering or phishing attacks against Dell customers.

Web3 Security Research

Ongoing

- Auditing smart contracts through Cantina and ImmuneFi.
- 4 submissions currently in review across multiple protocols with \$100M+ TVL, including Symbiotic and Term Finance.

Teaching Assistant | Software Architecture & Design

Aug 2025 – Dec 2025

- Mentored 50+ students on software-architecture principles and design patterns, running tutorials that grounded the theory in real-world scenarios, including constraints like limited internet access and remote-area connectivity.
- Improved grades and conceptual understanding across the class; students consistently pointed to the tutorials as what made the architecture-design material click, a clear and lasting impact.

Mindsquared | CTO (Remote)

Jan 2024 – Jun 2024

- Built data pipelines driving profitability roadmaps across Walmart and Amazon operations; helped 50+ companies reach profitability through automated analytics.
- Resolved 500+ customer complaints, closing the loop between engineering fixes and account-level support.

KEY PROJECTS

IntelliScore, AI Credit Scoring for Meezan Bank

Final Year Project, IBA Karachi

Led a four-person team from a one-line industry brief to a deployed, bank-facing AI product.

- Owned product direction for Meezan Bank (Pakistan's largest Islamic bank): scoped an undefined brief across six executive sessions, negotiated a data-sharing NDA, and absorbed three mid-flight scope pivots (consumer loans → MyBykea bike financing → auto financing) without slipping delivery, replacing a days-long manual ECIB review.
- Architected a semi-supervised teacher-student distillation pipeline over 93,000 anonymized records and 32 features (research-backed formula labeled 50k, a fine-tuned MLPRegressor 128-64-32 pseudo-labeled 30k, a HistGradientBoostingRegressor trained across both), cutting validation MSE from 15.67 to 3.12 at $R^2 = 0.9971$ and holding $R^2 = 0.99$ held-out at sub-2s inference.
- Built a RAG-grounded LLM explainability chatbot and shipped the full platform, UI and backend both in Python / Django, over PostgreSQL and Tailwind CSS, with multi-user auth and a comparative-analytics dashboard.

Attacking & Defending Confidential VMs

Computer Security, IBA Karachi

- Reproduced a published attack class (HECKLER): an untrusted hypervisor injects interrupts into hardware-isolated Confidential VMs on AMD SEV-SNP and Intel TDX TEEs, a gap open even when guest memory and CPU state are encrypted.
- Built a testbed on AMD EPYC (Milan/Genoa) across three iterations: stock QEMU/modified-kernel, a containerized build, then a patched-QEMU instance with a custom Python harness over the QEMU Machine Protocol (QMP); characterized guest-kernel behavior under injected NMI and

Machine-Check exceptions via forensic dmesg analysis.

- Proposed Guest_Defense, a guest-side mitigation validating that an interrupt matches the CPU's expected state at minimal overhead (advised by Dr. Faisal Iradat); separately reverse-engineered production Windows software (Camtasia, VEGAS, 7-Zip) to map control flow and licensing behavior.

SHA-256 Across Kernel, User Space & Syscalls

Operating Systems, IBA Karachi

- Built three separate custom Linux kernels (WSL) to isolate each path, implementing SHA-256 as a loadable kernel module via the kernel crypto API (crypto/hash.h) logging to dmesg, a user-space application over OpenSSL, and a custom system call exposing kernel cryptographic functionality to user programs.
- Benchmarked all three paths on identical inputs, quantifying the kernel/user-space boundary-crossing overhead and isolating where syscall and context-switch costs dominate.
- Analyzed the security trade-offs of each environment: privilege boundaries, access control, and the attack surface of running cryptographic code in ring 0 versus user space.

Nested Learning & Continual Adaptation

Neural Networks & Deep Learning, IBA Karachi

- Reproduced the core mechanisms of Nested Learning (2025), a reformulation of deep networks as a hierarchy of nested optimization processes at different time scales that recasts the optimizer as a memory system and targets catastrophic forgetting in deployed models.
- Implemented it from scratch in PyTorch (nested fast/medium/slow modules, a Continuum Memory System on EMA shadow weights, a self-modifying meta-learner that rewrites its own update rule each step), benchmarked against an MLP baseline to confirm optimization behaves as multi-rate memory.

Autonomous Maze-Solving Robot

Arduino & Discrete Logic, 3rd of 45 teams

- Built two independent from-scratch implementations of a maze-solving car on three IR sensors through a dual H-bridge driver: one in embedded C on Arduino, one as pure combinational hardware with no microcontroller and no firmware.
- Derived the full truth table and minimized it with Karnaugh maps into four Boolean equations in discrete logic ICs; the same equations drive the Arduino firmware unchanged, making the builds identical by construction. Placed 3rd of 45 teams.

AI Résumé Generator

Introduction to Text Analytics, IBA Karachi

- Fine-tuned Llama-2 7B with LoRA on 500 résumés, RAG-grounded for structure and tone; beat baseline ChatGPT head-to-head.

Real-Time E-Commerce Analytics Platform

Big Data Analytics, IBA Karachi

- Built an end-to-end streaming platform over 3-4 GB of high-velocity event data: ingest through Apache Kafka into a hybrid Hadoop/HBase and Redis layer, processing in Apache Spark, orchestrated as Apache Airflow DAGs and containerized in Docker across multiple nodes.
- Shipped three analytics workloads on one live pipeline (fraud detection, customer analytics, demand forecasting) via Python dashboards on Redis-cached aggregates, plus an operator dashboard exposing live ingestion throughput and HBase query telemetry.

CartNest, Full-Stack E-Commerce Platform

Web Application Development, IBA Karachi

- React SPA over Node.js/Express and PostgreSQL (Docker Compose); buyer/seller/admin roles on JWT + bcrypt, with Multer uploads.
- Clean routes/controllers/middleware with SQL migrations and Chart.js analytics.

Student Forum Portal, Full-Stack Community Platform

Database Systems, IBA Karachi

Stack: React, TypeScript, Node.js, SQLite

- Built a full-stack discussion forum (React/TypeScript over a Node.js/TypeScript + SQLite backend): threaded posts, comments, user accounts and content browsing end-to-end.
- Designed a normalized relational schema with enforced foreign-key constraints and indexed hot columns; used parameterized queries to eliminate SQL-injection risk and transactions for multi-step write consistency.

Parallel Matrix Convolution

Parallel & Distributed Computing, IBA Karachi

- Parallelized 2D matrix convolution (the core of image filtering) in C++ with OpenMP, decomposing the input into four quadrants across 8 concurrent threads to cut execution time over the serial baseline.
- Managed per-thread output aggregation and persisted the input, filter and result matrices to separate files, handling the partitioning and recombination that parallel decomposition requires.

NayaPay Wallet Redesign

Human-Computer Interaction, Figma

- 250 high-fidelity Figma screens on an 11-component design system across payment, billing and onboarding.
- Full Urdu RTL build (124 screens) mirroring the English set, covering 12 end-to-end flows.

SKILLS

Languages: Python, Java, JavaScript / TypeScript, C / C++, Assembly, SQL, Solidity, Verilog, Embedded C (Arduino)

Data Engineering: Apache Spark, Kafka, Airflow, Hadoop, HDFS, HBase, Hive, MongoDB, Redis, PostgreSQL, SQLite

Web & Tools: React, Node.js / Express, MERN / PERN stacks, Django, REST APIs, JWT auth, Tailwind CSS, Docker, Git, Linux, Figma

Security: Bug bounty, web app pentesting, Burp Suite, sqlmap, SQLi Dumper, Nmap, reverse engineering, binary analysis, QEMU, TEEs (SEV-SNP / TDX), CTF authoring, applied cryptography

Blockchain / Web3: Solidity, Ethereum, Foundry, smart-contract auditing & PoC development, contract verification, DApps, blockchain fundamentals

AI / ML: PyTorch, scikit-learn, neural networks, gradient boosting, teacher-student distillation, LoRA fine-tuning, RAG, LLMs (Llama-2, Qwen 2.5, Gemini API), CUDA C/C++

CERTIFICATIONS

NVIDIA Deep Learning Institute: Getting Started with Accelerated Computing in CUDA C/C++

2025

Cisco Networking Academy: Getting Started with Cisco Packet Tracer; Exploring Networking with Cisco Packet Tracer

2025

ACHIEVEMENTS & LEADERSHIP

- **Winner, Trading Bazaar**, IBA Karachi.
- **IBA Data Science Society:** Designed and authored the cybersecurity-track CTF challenges and judged submissions (validating solutions, scoring participants).
- **CryptoHack:** Completed the Introduction, General, Symmetric Ciphers, Diffie-Hellman, and Hash Functions categories.